

BCS603 / KCS603 — Computer Networks

Important Questions — Last 5 Years PYQ Analysis (2020–2024) | Unit-wise Breakdown

HOT = 3+ baar aaya

High Probability

NUM = Numerical Practice

Must Do (3+ times) High Importance (2 times) Medium Importance

TOTAL UNITS	TOTAL QS	MUST-DO	NUMERICALS	DIAGRAMS
5	45+	18	10	8

U1

Physical Layer & Network Basics

OSI · TCP/IP · Transmission Media · Modulation · Multiplexing · Switching

9 Qs

OSI model aur TCP/IP comparison GUARANTEED hai har saal. Diagram ke saath 7 layers yaad karo properly.

OSI Reference Model explain karo — 7 layers, unke functions, protocols aur services diagram ke saath. TCP/IP model se compare karo.

10M

EVERY YEAR

Bit Rate aur Baud Rate mein kya difference hai? Nyquist formula aur Shannon's theorem explain karo. Numerical: Channel capacity calculate karo.

2M

5M

REPEAT

NUMERICAL

Multiplexing kya hai aur uske types explain karo: FDM, TDM, WDM, STDM — working, diagram aur differences ke saath.

5M

10M

REPEAT

Circuit Switching aur Packet Switching mein kya difference hai? Advantages, disadvantages aur use cases ke saath compare karo. Message Switching bhi include karo.

5M

10M

REPEAT

Guided aur Unguided transmission media explain karo — Twisted pair, Coaxial, Fiber optic, Radio waves, Microwave, Infrared.

10M

5M

Modulation kya hai? Types explain karo: AM, FM, PM (Analog) aur ASK, FSK, PSK, QAM (Digital). Diagrams ke saath.

5M

10M

Point-to-Point aur Point-to-Multipoint mein difference explain karo. Network topologies (Star, Bus, Ring, Mesh) ke advantages/disadvantages.

2M

5M

2023 PYQ

- NRZ-L, NRZ-I, Manchester, Differential Manchester encoding schemes explain karo. Data 01001110 ke liye NRZ-L aur NRZ-I diagram draw karo.

5M

🔥 2022 PYQ

DIAGRAM

- Bandwidth aur Throughput mein kya difference hai? Latency, Propagation delay, Transmission delay define karo.

2M

U2

Data Link Layer

Framing · CRC · Hamming · ARQ · Sliding Window · ALOHA · CSMA

10 Qs

- 🔥 CRC numerical + Sliding Window Protocol numerical DONO guarantee hain. Stop-and-Wait efficiency formula yaad karo.

- CRC (Cyclic Redundancy Check) kya hai? CRC calculate karo:

Data: 1101011011, Divisor (Generator): 10011

Transmitted frame kya hogi? Error detection kaise hoti hai?

10M

🔥 EVERY YEAR

NUMERICAL

- Hamming Code kya hai? 7-bit received code **1010101** mein error detect karo aur correct karo. Even parity ke saath Hamming code construct karo.

10M

🔥 2022 PYQ

NUMERICAL

- Stop-and-Wait ARQ, Go-Back-N ARQ, aur Selective Repeat ARQ explain karo aur compare karo. Window size aur sequence number relationship explain karo.

10M

🔥 REPEAT

- Sliding Window Protocol numerical: $T_t = 1\text{ms}$, $T_p = 24\text{ms}$. Window size aur minimum sequence number find karo. GBN-3 mein 10 packets send karne hain, har 5th packet lost ho jaata hai — total transmissions?

10M

🔥 PYQ Confirmed

NUMERICAL

- Stop-and-Wait efficiency numerical: Agar efficiency kam se kam 50% chahiye, Transmission delay aur Propagation delay ka kya relationship hoga? Efficiency formula: $\eta = 1/(1+2a)$ derive karo.

5M

🔥 PYQ

NUMERICAL

- ALOHA protocols explain karo — Pure ALOHA aur Slotted ALOHA. Throughput formula, maximum efficiency aur difference. Slotted ALOHA 10% slots idle ho toh channels ka load kya hai?

5M

🔥 2022 PYQ

NUMERICAL

- CSMA/CD aur CSMA/CA explain karo — working, difference, aur kahan use hota hai (Ethernet vs WiFi). Carrier Sense protocols bhi explain karo.

5M

10M

- Piggybacking kya hota hai? Data Link Layer mein iska use kaise hota hai?

2M

🔥 2022 PYQ

- Framing techniques explain karo — Character count, Flag bytes with stuffing, Bit stuffing, Physical layer coding violations. Bit stuffing example solve karo.

5M

10M

- Network devices explain karo: Hub, Switch, Bridge, Router, Repeater — kaam, OSI layer, aur differences.

5M

🔥 2023 PYQ

U3

Network Layer

IP Addressing · Subnetting · Routing · ICMP · ARP · Congestion Control

10 Qs

🔥 Subnetting numerical GUARANTEED hai — Class B aur Class C dono practice karo. DVR/RIP aur Link State routing comparison bhi bahut aata hai.

- IP Addressing kya hai? Class A, B, C, D, E explain karo. Subnetting numerical: **Network 200.1.2.0 ko 5 subnets mein divide karo** — subnet mask, network address, broadcast address, host range.

10M

🔥 EVERY YEAR

NUMERICAL

- Subnet mask 255.255.192.0 diya hai aur IP 180.25.21.x — subnet address kya hoga? CIDR notation explain karo. Subnet mask se subnet count aur host count calculate karo.

2M

5M

🔥 2022 PYQ

NUMERICAL

- Distance Vector Routing (DVR/RIP) kya hai? Working example ke saath explain karo. Count-to-Infinity problem kya hai aur kaise solve hota hai?

10M

🔥 REPEAT

- Distance Vector Routing vs Link State Routing (OSPF) compare karo — 6+ features. Dijkstra's algorithm example solve karo given network graph.

10M

🔥 REPEAT

NUMERICAL

- ICMP (Internet Control Message Protocol) kya hai? Message types explain karo aur real-world scenarios mein use. BGP (Border Gateway Protocol) kya hai?

5M

🔥 2022 PYQ

- Congestion Control kya hai? Leaky Bucket aur Token Bucket algorithms explain karo — diagram aur difference ke saath. Congestion causes aur solutions.

10M

🔥 REPEAT

DIAGRAM

- ARP aur RARP kya hain? Working explain karo. ARP vs RARP vs DHCP kaise differ karte hain?

2M

5M

- IP Fragmentation kya hai? Fragment offset kaise calculate hota hai? Example: 4000-byte datagram, MTU=1500, fragments kaise banenge?

5M

NUMERICAL

- IPv4 aur IPv6 mein differences kya hain? IPv6 ke features explain karo.

5M

- Unicast, Multicast, aur Broadcast routing explain karo. Unicast routing protocols — RIP, OSPF, BGP — ek line difference.

5M

2M

U4

Transport Layer

TCP · UDP · 3-Way Handshake · Flow Control · Error Control · QoS

8 Qs

🔥 TCP vs UDP comparison aur TCP 3-way handshake GUARANTEED hai. TCP header format bhi draw karna aata ho.

- TCP aur UDP mein detailed comparison karo — header format, reliability, connection, use cases, speed. Kab TCP use hota hai aur kab UDP?

10M

🔥 EVERY YEAR

- TCP 3-way handshake (connection establishment) explain karo diagram ke saath: SYN → SYN-ACK → ACK. 4-way handshake (connection termination) bhi explain karo: FIN → ACK → FIN → ACK.

10M

🔥 REPEAT

DIAGRAM

- TCP Header format draw karo aur har field explain karo — Source Port, Dest Port, Seq No., ACK No., Window Size, Checksum etc.

10M

🔥 REPEAT

DIAGRAM

- Transport Layer kaise ensure karta hai ki complete message destination pe properly order mein pahunche? Flow control, error control, aur congestion control mechanisms explain karo.

10M

🔥 REPEAT

- UDP segment format draw karo aur fields explain karo. UDP kab use karna better hai? Real-time applications mein UDP kyun prefer hota hai?

5M

DIAGRAM

- Port addressing kya hai? Well-known ports, Registered ports, Dynamic ports explain karo. Examples: HTTP=80, HTTPS=443, FTP=21, SMTP=25, DNS=53.

2M

5M

- QoS (Quality of Service) parameters in transport layer explain karo — Reliability, Delay, Jitter, Bandwidth, Security.

5M

- TCP congestion control algorithms explain karo: Slow Start, Congestion Avoidance, Fast Retransmit, Fast Recovery.

5M

10M

U5

Application Layer & Network Security

DNS · FTP · SMTP · HTTP · RSA · Encryption · Digital Signatures · Firewall

9 Qs

🔥 FTP + SMTP + DNS — teeno ek saath 10M mein aate hain. RSA algorithm numerical bhi frequently aata hai — practice karo.

- FTP, SMTP, aur DNS explain karo — working, protocol type, port numbers, use cases. Ye question 10M mein combine aata hai: "Explain: (i) FTP (ii) SMTP (iii) DNS (iv) ARP"

10M

🔥 2022+2023 PYQ

- DNS (Domain Name System) resolution process explain karo — Recursive query, Iterative query, DNS hierarchy (Root → TLD → Authoritative). DNS records types: A, MX, CNAME, NS.

10M 🔥 REPEAT DIAGRAM
- RSA algorithm explain karo aur numerical solve karo:
p=7, q=11 → n=77, $\phi(n)$ =60 → public key e=17 → private key d=?
Message M=5 encrypt karo aur decrypt karo.

10M 🔥 REPEAT NUMERICAL
- Symmetric aur Asymmetric encryption mein kya difference hai? RSA, DES, AES ke naam aur use. Digital Signatures kya hote hain aur kaise kaam karte hain?

5M 10M 🔥 REPEAT
- HTTP aur HTTPS explain karo — message structure, request/response format, GET vs POST methods, status codes (200, 404, 500). HTTP 1.0 vs 1.1 vs 2.0 differences.

5M 10M 🔥 REPEAT
- SMTP ke 3 phases explain karo — Connection establishment, Mail transfer, Connection termination. SMTP vs POP3 vs IMAP difference.

5M
- Firewall kya hota hai? Types of firewalls explain karo — Packet filtering, Stateful inspection, Application gateway. Network security ke threats bhi explain karo.

5M 10M
- FTP ka working detail mein explain karo — Control connection (port 21) aur Data connection (port 20). Active mode vs Passive mode.

5M
- Networking commands explain karo: ipconfig, netstat, ping, hostname, tracert — kya karta hai har command aur kab use hota hai.

5M 🔥 PYQ

 Exam Strategy — Section-wise Breakdown

SECTION A – 2M × 10 = 20 Marks (Must attempt all)

- ✓ Bit rate vs Baud rate
- ✓ Piggybacking define karo
- ✓ Count-to-infinity problem
- ✓ Point-to-point vs Point-to-multipoint
- ✓ Subnet address calculate (180.25.21.x, mask 255.255.192.0)
- ✓ ICMP role define karo
- ✓ Port addressing (well-known ports)
- ✓ ARP/RARP define karo
- ✓ Symmetric vs Asymmetric (one liner)
- ✓ QoS parameters list karo

SECTION B/C/D/E/F – 10M × 5 = 50 Marks

- ✓ 🔥 OSI Model (7 layers + diagram)
- ✓ 🔥 CRC Numerical (1101011011 ÷ 10011)
- ✓ 🔥 Sliding Window / GBN Numerical
- ✓ 🔥 Subnetting Numerical (200.1.2.0 → 5 subnets)
- ✓ 🔥 DVR/Dijkstra routing numerical
- ✓ 🔥 TCP vs UDP (comparison table)
- ✓ 🔥 TCP 3-way handshake
- ✓ 🔥 FTP + SMTP + DNS combined
- ✓ 🔥 RSA Numerical (p=7, q=11)
- ✓ 🔥 Leaky Bucket / Token Bucket